

VIKRAM ADITYA RAJESH

AI/ML Engineer

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Machine Learning Engineer specializing in production-grade AI systems, enterprise backend development, and custom RAG pipelines. Experienced in building scalable applications using React, Spring Boot, Azure cloud services, FAISS, and LLM-based workflows. Strong background in document intelligence, retrieval optimization, REST APIs, and deploying AI-powered enterprise solutions for real-world use cases.

EXPERIENCE

Thoughtfocus

Jan 2025 - Present

AI Engineer

- Built an internal HR Q&A system using a RAG architecture with semantic chunking, all-MiniLM-L6-v2 embeddings, FAISS, and a locally fine-tuned Qwen 3 model deployed through Ollama, enabling low-latency retrieval over 200+ policy documents and structured employee data with role-based access control and workflow-aware retrieval. Fine-tuned the model using Unsloth and QLoRA for improved domain-specific HR query understanding and response quality.
- Built Selenium based automation for Adobe Experience Manager (AEM) forms using Faker generated dynamic datasets to automate end-to-end form validation, reducing manual regression testing effort by 60%.
- Mentored an intern during an internal AI project, guiding implementation and retrieval pipeline development to improve delivery quality and development speed.
- Developed PitStop Manager, an internal enterprise HR and recruitment platform using React, Spring Boot REST APIs, Azure Container Apps, Azure PostgreSQL, and Azure Blob Storage, with Azure AD SSO authentication and semantic skill similarity matching between resumes and job descriptions for candidate screening and workflow automation.

Tech Mahindra

Nov 2024 - Jan 2025

Internship

- Contributed to Project Indus, Tech Mahindra's enterprise LLM initiative, by developing and evaluating RAG workflows, benchmarking LLM response quality across accuracy and relevance metrics, and implementing experimentation pipelines using Hugging Face Transformers, PEFT, TRL, and bitsandbytes on GPU-backed GCP environments.
- Built and integrated backend APIs using Django and MySQL for an internship application platform, enabling seamless communication between frontend interfaces and server-side systems.

Mavencart

June 2023 - Aug 2023

Internship

- Contributed to backend development for AI products, including a generative AI writer and social media template generator, supporting scalable content generation workflows and reducing manual content creation effort.

PROJECTS

Image Colorization [GitHub]

Python, OpenCV, Flask

Implemented OpenCV DNN inference using a pretrained Caffe colorization model to predict ab channels in LAB color space and reconstruct colorized images from grayscale inputs, with a Flask web interface for real-time image colorization.

HR Q&A system [Internal]

Python, FAISS, Sentence Transformers, Unsloth, QLoRA, Streamlit

Built a document aware AI assistant for querying enterprise PDFs and structured data using semantic retrieval pipelines and a locally finetuned Qwen 3 model deployed through Ollama for domain specific response generation.

EDUCATION

MVJ College of Engineering

Bangalore, Karnataka

Bachelor of Engineering in Artificial Intelligence and Machine Learning

2020 - 2024

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C++

ML Frameworks: TensorFlow, Keras, PEFT, TRL, Unsloth, QLoRA, FAISS, Sentence Transformers, Ollama

Backend & Web: Spring Boot, FastAPI, Django, React, Node.js, REST APIs, Streamlit

Cloud & Database: Azure(Container Apps, Blob Storage, Active Directory, Application Insights, PostgreSQL), MySQL

Libraries & Automation: NumPy, pandas, scikit-learn, Matplotlib, OpenCV, NLTK, Selenium, Playwright, Faker